

HOTEL BOOKING DEMAND ANALYSIS

Hypothesis Testing Report

Project : Hotel Booking Demand Analytics Platform

Tools : Python, SciPy, Statsmodels, Pandas

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Data Analytics Project

Project Overview

Hypothesis testing was performed to determine whether observed differences and relationships in hotel booking data are statistically significant. Both parametric and non-parametric statistical tests were applied after validating statistical assumptions. The results provide evidence-based conclusions to support hotel pricing strategies, customer analysis and revenue management.

Independent t-Test

Objective

The Independent t-Test was performed to determine whether the Average Daily Rate (ADR) differs significantly between City Hotels and Resort Hotels.

Null Hypothesis (H₀)

There is no significant difference in the Average Daily Rate (ADR) between City Hotels and Resort Hotels.

Alternative Hypothesis (H_a)

There is a significant difference in the Average Daily Rate (ADR) between City Hotels and Resort Hotels.

Statistic	Value
Test	Independent t-Test
Variable	ADR
t Statistic	30.287
p-value	6.153245924852696e-200
Significance Level	0.05
Decision	Reject H ₀

Business Interpretation

The p-value is less than 0.05, therefore the null hypothesis is rejected. This indicates that the Average Daily Rate (ADR) differs significantly between City Hotels and Resort Hotels. Hotel managers should consider separate pricing strategies for different hotel types to maximize occupancy and revenue.

One-Way ANOVA

Objective

One-Way Analysis of Variance (ANOVA) was performed to determine whether the Average Daily Rate (ADR) differs significantly among different customer types.

Null Hypothesis (H₀)

There is no significant difference in the Average Daily Rate (ADR) among different customer types.

Alternative Hypothesis (H_a)

At least one customer type has a significantly different Average Daily Rate (ADR).

Statistic	Value
Test	One-Way ANOVA
Variable	ADR
F Statistic	669.1282713505746
p-value	0.000
Alpha	0.05
Decision	Reject H ₀

Business Interpretation

Since the p-value is less than 0.05, the null hypothesis is rejected. This indicates that customer type significantly influences the Average Daily Rate (ADR). Hotels should design pricing strategies based on customer segments to improve occupancy and maximize revenue.

Chi-Square Test

Objective

The Chi-Square Test of Independence was performed to determine whether there is a significant association between Hotel Type and Booking Cancellation.

Null Hypothesis (H_0)

Hotel Type and Booking Cancellation are independent. There is no significant association between them.

Alternative Hypothesis (H_a)

Hotel Type and Booking Cancellation are associated. There is a significant relationship between them.

Statistic	Value
Test	Chi-Square
Variables	Hotel vs Cancellation
Chi-Square Statistic	245.68
p-value	6.905961865776799e-101
Alpha	0.05
Decision	Reject H_0

Business Interpretation

Since the p-value is less than the significance level (0.05), the null hypothesis is rejected. This indicates a statistically significant association between Hotel Type and Booking Cancellation. The cancellation behaviour differs across hotel types, suggesting that cancellation reduction strategies should be designed separately for City Hotels and Resort Hotels.

Mann-Whitney U Test

Objective

The Mann-Whitney U Test was performed to compare the distribution of Average Daily Rate (ADR) between City Hotels and Resort Hotels. Since the data did not satisfy the normality assumption, this non-parametric test was selected as an alternative to the Independent t-Test.

Null Hypothesis (H₀)

There is no significant difference in the distribution of Average Daily Rate (ADR) between City Hotels and Resort Hotels.

Alternative Hypothesis (H_a)

There is a significant difference in the distribution of Average Daily Rate (ADR) between City Hotels and Resort Hotels.

Statistic	Value
Test	Mann-Whitney U Test
Variable	ADR
U Statistic	1135645990.0
p-value	0.0
Alpha	0.05
Decision	Reject H ₀

Business Interpretation

The Mann-Whitney U Test provides a robust comparison of ADR between hotel types without assuming normality. The test result indicates whether the booking revenue distribution differs significantly between City Hotels and Resort Hotels. These findings help hotel managers design pricing and marketing strategies tailored to each hotel category.

Kruskal-Wallis Test

Objective

The Kruskal-Wallis Test was performed to determine whether there are significant differences in ADR among different customer groups without assuming normality.

Null Hypothesis (H₀)

No significant difference exists among customer groups.

Alternative Hypothesis (H_a)

At least one customer group differs significantly.

Statistic	Value
Test	Kruskal-Wallis
H Statistic	2478.85
p-value	0.0
Decision	Reject H ₀

The Kruskal-Wallis test confirms whether booking revenue differs significantly across customer groups, helping management identify high-value customer segments.

Pearson Correlation

Pearson Correlation measures the strength of linear relationships between numerical variables.

Statistic	Value
Variables	Lead Time vs ADR
Correlation	0.0219
p-value	8.997006162842638e-11
Decision	Significant

Pearson correlation helps identify whether two numerical variables increase or decrease together.

Spearman Correlation

Spearman Correlation measures monotonic relationships between variables without assuming normality.

Statistic	Value
Variables	Lead Time vs ADR
Correlation	0.105
p-value	7.675400161676427e-215
Decision	Significant

Spearman correlation is useful for identifying monotonic relationships when data is not normally distributed.

Overall Statistical Decisions

• Independent t-Test completed • Chi-Square Test completed • Mann-Whitney U Test completed • Kruskal-Wallis Test completed • Pearson Correlation completed • Spearman Correlation completed The statistical analysis indicates significant relationships among booking behaviour, hotel type, customer characteristics and revenue patterns.

Business Recommendations

• Improve pricing based on hotel type. • Monitor cancellation behaviour regularly. • Develop customer-specific marketing strategies. • Use statistical insights for occupancy forecasting. • Build predictive models for future booking analysis. • Continue collecting quality customer data.

Conclusion

Hypothesis testing confirmed statistically significant relationships within the Hotel Booking Demand dataset. The results support data-driven decision making for pricing, customer segmentation, revenue optimization and hotel management. This report concludes the statistical validation phase of the Hotel Booking Demand Analytics Platform.